

- 5 (a) transducer/transmitter can be also be used as the receiver
or
transducer both transmits and receives

receives reflected pulses between the emitted pulses

(needs to be pulsed) in order to measure/determine depth(s)

(needs to be pulsed) to determine nature of boundaries

Any three of the above marking points, 1 mark each

B2 [2]

- (b) (i) product of speed of (ultra)sound and density (of medium)

M1

reference to speed of sound in medium

A1 [2]

- (ii) if Z_1 and Z_2 are (nearly) equal, I_T/I_0 (nearly) equal to 1/unity/(very) little reflection/mostly transmission

B1

if $Z_1 \gg Z_2$ or $Z_1 \ll Z_2$ or the difference between Z_1 and Z_2 is (very) large, then I_T/I_0 is small/zero/mostly reflection/little transmission

B1 [2]